

Mid Logic

Q1.

choose the correct answer :

#Simplified form of the logic expression $(A + B' + C)(A + B' + C')(A + B + C)$

a) $A'B + C'$

b) $A + B'C$

c) A

d) $AB + C'$

#The minimal POS function $G(A,B,C,D)$ which represent by following K- map is?

0	1	1	0
1	1	1	1
1	1	0	0
0	0	0	0

a) $G(A,B,C,D) = (A' + B)(B + D)(A' + C')$

b) $G(A,B,C,D) = A'D + BC' + A'B$

c) $G(A,B,C,D) = M(0,2,8,9,10,11,14,15)$

d) $G(A,B,C,D) = (A + B')(B' + D')(A + C)$

#Binary data is being represented in size of byte and in 2's complement form. The number of 0's in the representation of (-127) decimal is?

A) 8

B) 7

C) 6

D) 5

#For a binary half-subtractor having two inputs A and B, the correct set of logical expressions for the outputs D (= A minus B) and X (= borrow) are?

- A. $D=AB+A'B$, $X= A'B$
- B. $D=A'B+ AB' + AB$, $X= AB'$
- C. $D=A'B + AB'$, $X=A'B$
- D. $D=AB+A'B'$, $X=AB'$

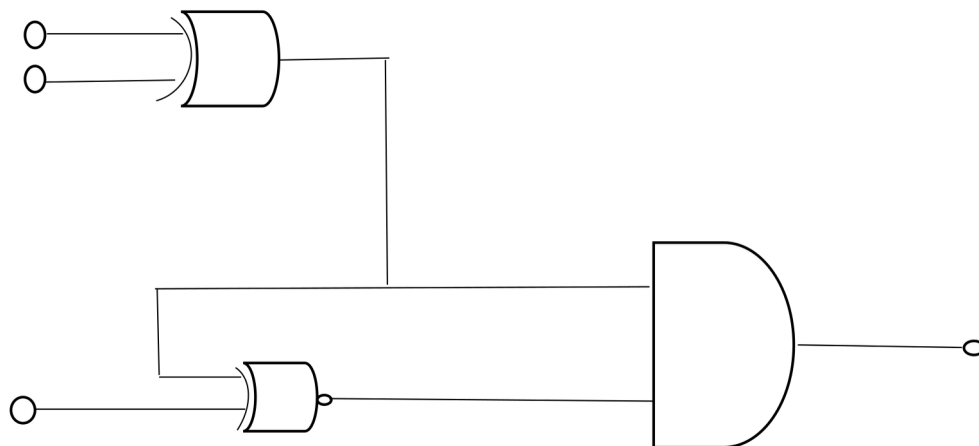
The minterm expansion of $f(P,Q,R) = PQ + QR' + PR'$ is?

- A) $m_2+m_4+m_6+m_7$
- B) $m_0+m_1+m_3+m_5$
- C) $m_2+m_3+m_4+m_5$
- D) $m_0+m_1+m_6+m_7$

#Consider minterm expansion of the function $F(P, Q,R,S) = (0,5, 10,15)$ and $F(P,Q,R,S)$ don't care $= (2, 7,8, 13)$ The minimal sum of product form of F is?

- A. $QS'+Q'S$
- B. $Q'RS +QR'S + QRS' + Q'R'S$
- c. $Q'S' +QS$
- D. $P'QS+PQ'S+PQS'+P'Q'S$

#The function F=



- a) $XY'Z + X'YZ$

B) $X'Y'Z + X'YZ'$

C) $X'Y'Z' + XYZ$

D) $XY'Z'$

#Minimized expression for $(x + y) + \overline{(x + y)} + \overline{(XY' + X')}$ is

A) X

B) XY

C) Y

D) X+Y

Q2.

Fill the box with the correct answer:

Convert the Gray code number (11010) to decimal number.

Convert the Gray number 110011 to excess-3.

Convert the octal number (25) to excess-3.

Convert the Gray number 110011 to Hexa-decimal.

On addition of 46 and +28 using 2's complement, we get

What is the addition of the binary numbers 11011011010 and 010100101

Q3.

choose the correct answer:

#The code where all successive numbers differ from preceding number by a single bit is

A. ASCII Code

B. BCD

C. Excess 3

D. Gray

#Convert the following SOP expression to an equivalent POS expression

$ABC + AB'C' + ABC + ABC + A'B'C$

A. $(A'+B'+C')(A+B+C)(A+B+C')(A'+B'+C)(A+B+C)$

B. $(A+B+C)(A+B+C)(A+B+C)$

C. $(A+B+C')(A+B+C)(A+B+C)$

D. $(A+B+C)(A+B'+C')(A+B+C)$

#The Minimum number of 2-input NAND gates required to implement the function $F=(x+y)(Z+W)$ is

A. 3

B. 4

C. 5

D. 6

#For the K map of the given figure, the simplified Boolean expression is

A. $A'C'+A'D'+ABC$

B. $AB'D+BC$

C. $AC'D+AC$

D. $AC'D+AC+BC$

1	1		1
1	1		1
		1	1

#The number of product terms in the minimized sum of product expression obtained through the following K-map, where d, don't care is:

A. 2

B. 3

C. 4

D. 5

1	0	0	1
0	d	0	0
0	0	d	1
1	0	0	1

#Minimize the following expression using K-map $Y=m(0,1,5,9,13,14,15)+d(3,4,7,10,11)$

A. $D+A'C'+A'C$

B. $C+AD+A'D'$

C. $D+A'C'+AB$

D. $D+AC+A'C'$

#How many bits are needed to store one BCD digit?

A. 2 bits

B. 4 bits

C. 3 bits

D. 1 bit

#Convert $(312)_8$ into decimal.

A. $(201)_{10}$

B. $(202)_{10}$

C. $(203)_{10}$

D. $(204)_{10}$

#What is the addition of the binary number $101001 + 10011 =$

A. 010100

B. 111100

C. 000111

D. 101110

#The following hexadecimal number $(1E.43)_{16}$ is equivalent to

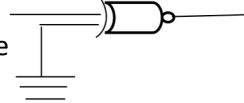
A. $(36.506)_8$

B. $(35.506)_8$

C. $(36.206)_8$

D. $(35.206)_8$

#For the gate in the given figure the output will be



A. 0

B. 1

C. A'

D. A

#The parity is used to

A. Detect errors

B. Convert to digital

C. Convert to analog

D. B and c

#The BCD for $(17)_{10}$ is

A. 00101111

B. 001111

C. 10001

D. 00010111

#Convert $(10111.0101)_2$ Into decimal.

A 23.3125

B 23.25

C 23.5

D. 373

#the complement of $f = x' + yz' + xy'z$ is

A. $x' + (y' + z)(x' + y + z')$

B. $x' (y + z')(x + y' + z)$

C. $x (y' + z)(x' + y + z')$

D. $x + y' z + x'y z'$

Q4. Perform the subtraction on the given numbers using the 2's complement $(1110)_2 = (10001)_2$

Q5. simplify the following Boolean expression to minimum number of literals $A'B(D'+C'D)+B(A+A'CD)$

Q6.

choose the correct answer:

#What is the binary equivalent of the decimal number 368

A. 101110000

C. 111010000

B. 110110000

D. 111100000

#The simplification of the Boolean expression is

$F = A'BC' + AB'C'$

A. A

B. 1

C. 0

D. BC

#The Gray code for decimal number 6 is equivalent to

A. 1100

B. 0101

C. 1001

D. 0110

#The Boolean expression $A'.B + A.B' + A.B$ is equivalent to

A. $A+B$

B. $A'.B$

C. $A+B'$

D. AB

#The output of a logic gate is 1 when all its inputs are at logic 0, the gate is either

A. a NAND or an EX-OR

B. an OR or an EX-NOR

C. an AND or an EX-OR

D. a NOR or an EX-NOR

Q7. Perform the subtraction on the given numbers using the 2's complement $(1110)_2 - (10001)_2$

Q8. Derive the circuits for a three-bit parity generator using an odd parity bit with showing the truth table for the parity generator.

Q9. Simplify and draw the logic diagram for the given expression. $F = (ABC)' + AB(C' + A'BC' + A)BC' + AB'C$.

For the given Boolean function $F(A,B,C,D) = A'B'C' + AC'D' + BC$

Which has the don't care condition $d(A,B,C,D) = \Sigma(2,3,4,10,11)$

b. Minimize f as product-of-sum (without don't care)

a. Draw the K_map for function f

c Minimize f as sum-of-products (with don't care)

d. convert and draw the minimized f in branch (c) to NAND gate only circuit

Q.1)

1	2	3	4	5	6	7	8
B	a	C	C	A	C	a	b

Q.2)

1) = 19

2) = ~~100010~~ = ~~(34)₁₆~~ = 0111 0110

3) = 0100 0101

4) = (22)₁₆

5) = 01001010

6) = 1110111111

Q.3)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
D	a	B	C	a	a	B	B	B	a	C	a	D	a	D	a	C

Q.4) = (11101)₂ = (-3)₁₀

Q.5) F = B

Q.6)

1	2	3	4	5
a	B	B	a	d

Q. 8) $P = (A \oplus B \oplus C)'$

Q. 9) $f = A' + B' + C'$

